On The Way to Temporal OBDA Systems

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Ontology-Based Data Access

- Extend data with external knowledge (ontology)
  Some examples:
  - every *father* is a *parent*
  - every paper must have some author
- This knowledge is made available through rewriting
  - Often referred to as a *virtual knowledge graph*
- Many commercial and open-source systems:
  Mastro, Stardog, Ontop, Morph, …

Temporal Data

- Temporal databases are not new ...[Snodgrass and Ahn, '86]
- ...but we see a recent rise in new temporal database systems:
  - InfluxDB,
  - Prometheus,
  - TimeScaleDB, and many others.
- User access to complex relations in temporal data is a challenge; systems generally do not allow rich, *declarative* queries over time
- This naturally led to the idea of solving this issue via OBDA.

Myriads of Proposals

- There has been a lot of work done on both theoretical questions of temporal OBDA, exploring questions of decidability and computational complexity
- Also many proposals for specific temporal rule and query languages.
- There have even been proposals for specific temporal OBDA systems, as recently as 2019.
- However: so far there has not been any progress on realising these ideas.

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Our Contribution

1. Identify the challenges that need to be overcome to achieve temporal OBDA in practice

2. Develop a realistic model for OBDA

3. Work towards implementing it.

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   - interval-based relations
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Project

time	*budget*
(0, 15)	150
(16, 30)	300

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3. **Expressive, Composable Query Language.**
   - Many existing proposals for this (DslD,datalogMTL).
   - But we also want to output temporal relations again, not just intervals or time points.

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- For this model, we take the temporal query language from [Klarmann and Meyer, RR ’14] as a basis.
- The ontology is based on DL Lite, but allows for temporal ABox.
- Mapping temporal relations (both time-series and interval-based) to the temporal ABox.
- Temporal reasoning on the query level: extending CQs via existential quantification over interval variables.
  - Aiming for rewritability to FO($\leq$), i.e. into SQL ultimately.
- Ongoing work to implement this on the basis of Ontop

Conclusion & Future Work

- Begun to formalise the challenges that temporal OBDA systems need to meet.
- Currently working on a first prototype system implemented on the basis of Ontop.
- For future we envision practical systems that meet all the challenges outlined here.
  - In addition to this, we are interested in exploring aggregation, richer semantics on intervals (duration function) as well as exploring ways to introduce new temporal events into the ontology level.

Thank You For Your Attention!

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